

# Raman analysis — Spec0093 — Clean [CDAE+CBRAE] — CR-G3-05\_25022026

*Fit + RRUFF match analysis report*

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|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>Sample :</b>      | CR-G3-05\25022026\Spectrum_93.txt                 |
| <b>Fit :</b>         | Spec0093 — Clean [CDAE+CBRAE] — CR-G3-05_25022026 |
| <b>Source :</b>      | Clean                                             |
| <b>Date :</b>        | 2026-05-05 00:11                                  |
| <b>Operator :</b>    | Ir. Basile-Valdimir Fauconnier                    |
| <b>Email :</b>       | bvfauconnier@gmail.com                            |
| <b>Source file :</b> | Raw_Spectrum\CR-G3-05\25022026\Spectrum_93.txt    |

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# 1. Spectrum preprocessing

The raw spectrum underwent 2 preprocessing step(s) before fitting, applied in the following order:

- **Step 1 (CDAE)** : Denoising via convolutional autoencoder (Convolutional Denoising AutoEncoder)
- **Step 2 (CBRAE)** : Baseline correction via convolutional autoencoder (Convolutional Baseline Removal AutoEncoder)

Technical trace : [CDAE+CBRAE]

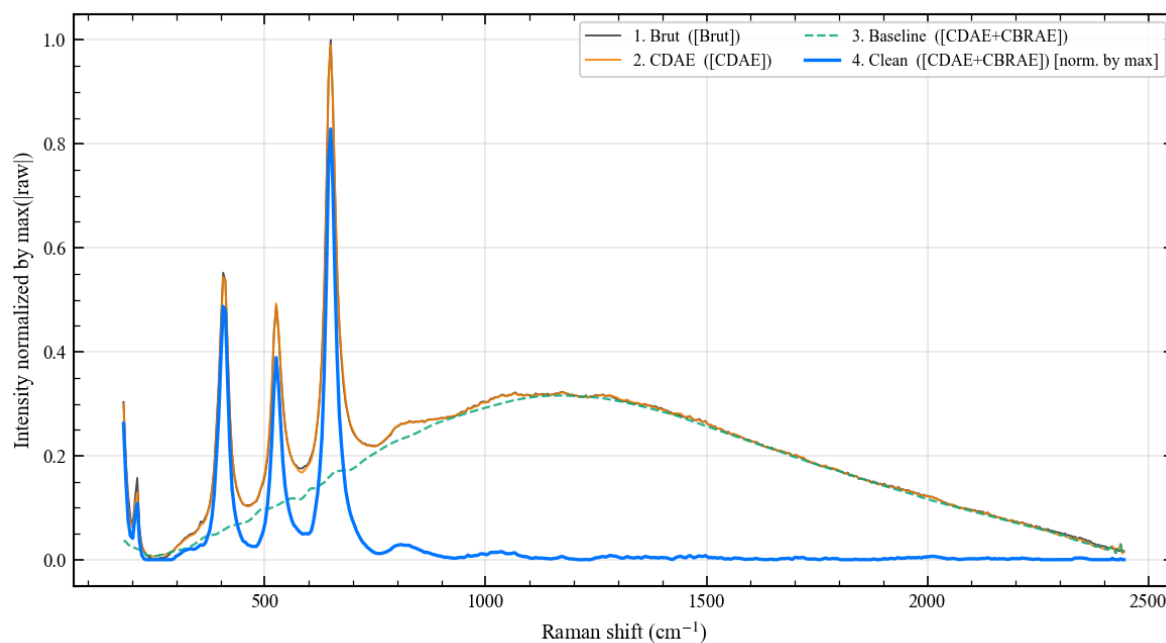


Figure: Overlay of the preprocessing steps applied to the spectrum — from raw (gray, in background) to the final state used for the fit (source color, thick line in foreground). Each curve is normalized by its absolute maximum intensity to allow comparison of spectral shapes.

## 2. Spectrum & fit

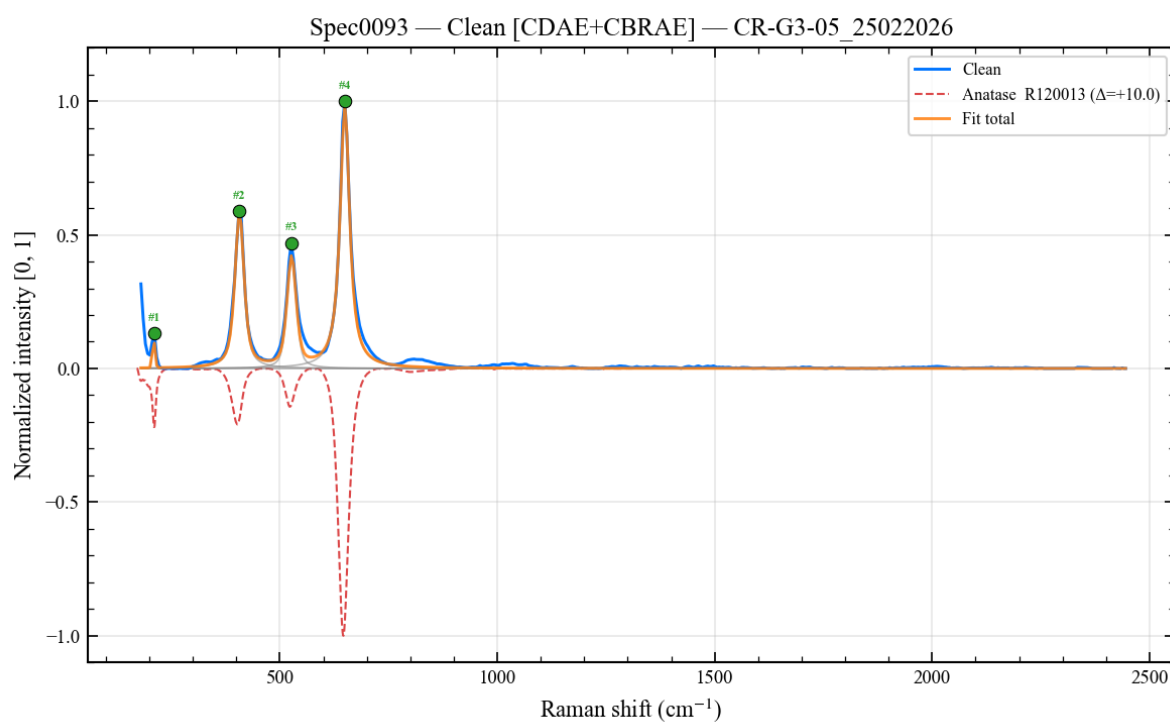


Figure 2: Source spectrum, fitted peak profiles and total fit. Colored markers indicate detected peaks (green = matched, red = unmatched if match analysis available).

### 3. Detected and fitted peaks

Total: 4 peak(s) detected, 4 fitted successfully.

| # | Position ± err<br>(cm <sup>-1</sup> ) | FWHM ± err<br>(cm <sup>-1</sup> ) | Area  | R <sup>2</sup> | Shape       | η    | Origin |
|---|---------------------------------------|-----------------------------------|-------|----------------|-------------|------|--------|
| 1 | 210.01 ± 1.12                         | 7.20 ± 4.01                       | 0.818 | 0.9333         | pseudovoigt | 0.00 | auto   |
| 2 | 407.28 ± 0.17                         | 24.85 ± 0.58                      | 20.3  | 0.9958         | pseudovoigt | 0.68 | auto   |
| 3 | 527.08 ± 0.34                         | 25.10 ± 1.11                      | 13.5  | 0.9824         | pseudovoigt | 0.50 | auto   |
| 4 | 649.51 ± 0.20                         | 25.95 ± 0.77                      | 39.4  | 0.9947         | pseudovoigt | 0.94 | auto   |

R<sup>2</sup> global du fit total = 0.9652

## 4. Automatic identification

**Method.** The automatic scan evaluates the correspondence between the fitted peaks of the studied spectrum and each RRUFF spectrum in the active subfolder. For each reference, an optimal spectral offset  $\Delta$  is searched by sweep ( $\Delta \in [-25, +25] \text{ cm}^{-1}$ , step 0.5) to maximize the weighted F1 score. The matching tolerance for a peak is adaptive:  $\max(5 \text{ cm}^{-1}, 3 \cdot \sigma_{\text{fit}})$  where  $\sigma_{\text{fit}}$  is the uncertainty of the Gauss/Lorentz/Pseudo-Voigt fit on the peak position.

**Metrics.** **M1 (coverage)** = fraction of the studied spectrum's peaks found in the reference. **M2 (signature)** = fraction of RRUFF peaks found in the studied spectrum. **M3 (F1-score)** = harmonic mean of M1 and M2, main ranking metric. The candidate with the highest M3 is considered the best match.

Scan date: 2026-05-05 00:11:06  
Files scanned: 11415/11415  
RRUFF folder: DATABASE\_RRUFF\excellent\_unoriented  
Candidates kept: 1955

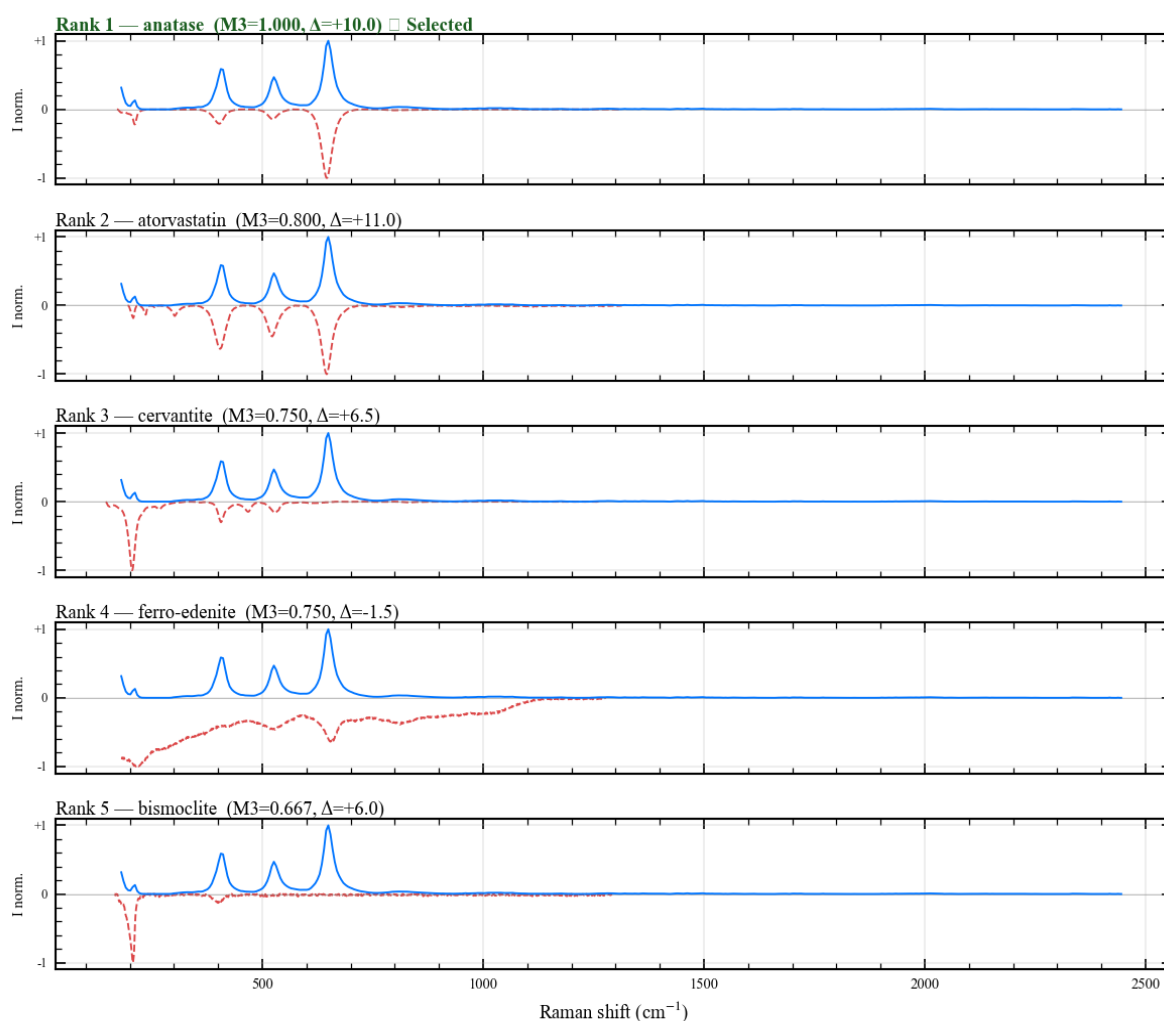


Figure: Visual comparison of the Top 5 identified candidates. The source spectrum (blue) is plotted above axis 0; each candidate (red dashed) is plotted mirrored with its optimal offset  $\Delta$  applied. The ★ indicates the mineral kept for detailed match analysis.

### Top 10 candidates (ranked by M3):

| # | Mineral | Formula          | M3    | M1   | M2   | $\Delta v$<br>mean<br>( $\text{cm}^{-1}$ ) | $\Delta$ auto<br>( $\text{cm}^{-1}$ ) | Match/<br>Fit | Choice     |
|---|---------|------------------|-------|------|------|--------------------------------------------|---------------------------------------|---------------|------------|
| 1 | anatase | TiO <sub>2</sub> | 1.000 | 1.00 | 1.00 | -2.9                                       | +10.0                                 | 4/4           | ★ Selected |

|    |                   |  |       |      |      |      |       |     |  |
|----|-------------------|--|-------|------|------|------|-------|-----|--|
| 2  | atorvastatin      |  | 0.800 | 1.00 | 0.67 | -3.5 | +11.0 | 4/4 |  |
| 3  | cervantite        |  | 0.750 | 0.75 | 0.75 | -1.4 | +6.5  | 3/4 |  |
| 4  | ferro-edenite     |  | 0.750 | 0.75 | 0.75 | +3.1 | -1.5  | 3/4 |  |
| 5  | bismoclite        |  | 0.667 | 0.50 | 1.00 | -3.8 | +6.0  | 2/4 |  |
| 6  | gahnite           |  | 0.667 | 0.50 | 1.00 | +3.9 | -4.5  | 2/4 |  |
| 7  | pargasite         |  | 0.667 | 0.50 | 1.00 | +0.2 | -24.5 | 2/4 |  |
| 8  | rynersonite       |  | 0.667 | 0.50 | 1.00 | +3.2 | -21.0 | 2/4 |  |
| 9  | schwartzembergite |  | 0.667 | 0.50 | 1.00 | +0.6 | -12.0 | 2/4 |  |
| 10 | senarmontite      |  | 0.667 | 0.50 | 1.00 | -1.4 | +11.0 | 2/4 |  |

**Discussion.** The top-ranked mineral is **anatase** ( $M3 = 1.000$ , optimal  $\Delta = +10.0 \text{ cm}^{-1}$ ). The score is moderately higher than the followers, identification is plausible but visual verification is recommended (relative gap to rank 2 = 20.0%). The rank 1 mineral was selected for the detailed match analysis presented in the next section.

## 5. RRUFF match analysis

**Active reference:** Anatase R120013  
**RRUFF file:** Anatase\_R120013\_Raman\_532\_\_\_\_\_Raman\_Data\_Processed\_\_4ecb65ee10af8685a3f304966693.txt  
**Laser:** 532 nm  
**Offset  $\Delta$  used:** +10.00  $\text{cm}^{-1}$   
**Formula:**  $\text{TiO}_2$   
**Group:** Ti oxides

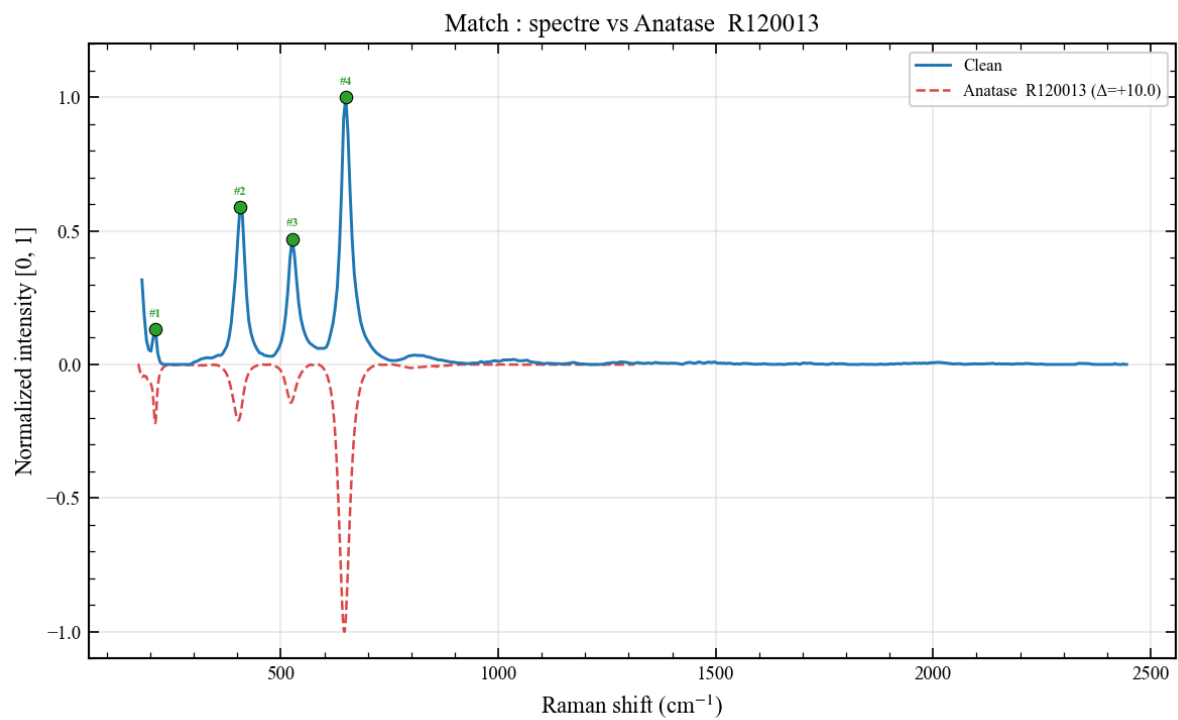


Figure 2: Studied spectrum vs RRUFF reference comparison (with applied offset). Matched peaks in green, unmatched in red, unobserved RRUFF peaks in dashed purple.

### Similarity scores

| Metric               | Value                  | Interpretation                      |
|----------------------|------------------------|-------------------------------------|
| M1 — Coverage        | 1.000                  | 4/4 spectrum peaks matched          |
| M2 — Signature       | 1.000                  | 4/4 RRUFF peaks recovered           |
| M3 — F1-score        | 1.000                  | Harmonic combination $M1 \times M2$ |
| $\Delta v$ mean      | -2.90 $\text{cm}^{-1}$ | Residual systematic shift           |
| $\Delta v$ std. dev. | 2.27 $\text{cm}^{-1}$  | Dispersion of shifts                |

## 6. Peak-to-peak correspondences

| # | Spectrum peak<br>(cm <sup>-1</sup> ) | RRUFF peak<br>(cm <sup>-1</sup> ) | $\Delta\nu$<br>(cm <sup>-1</sup> ) | Tol. | Status    | Mode                            |
|---|--------------------------------------|-----------------------------------|------------------------------------|------|-----------|---------------------------------|
| 1 | 210.01                               | 210.90                            | +0.89                              | ±5.0 | ✓ MATCHED | E <sub>g</sub> <sup>(2)</sup>   |
| 4 | 649.51                               | 646.25                            | -3.26                              | ±5.0 | ✓ MATCHED | ★ E <sub>g</sub> <sup>(3)</sup> |
| 3 | 527.08                               | 522.83                            | -4.25                              | ±5.0 | ✓ MATCHED |                                 |
| 2 | 407.28                               | 402.30                            | -4.98                              | ±5.0 | ✓ MATCHED |                                 |